

COURSE SYLLABUS

Advanced Automated Administration with Windows PowerShell

COURSE NUMBER

65319A

DURATION

3 days · 12 modules

AUDIENCE LEVEL

Advanced

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This 3-day course teaches students how to automate administrative tasks using PowerShell on Windows Server 2025. Students will learn essential scripting skills and advanced automation techniques for Microsoft Azure, Active Directory, Entra ID, SQL Server 2025, virtual machines, remote administration, and file management. Special attention is focused on new capabilities introduced in Windows Server 2025 and PowerShell 7+.

AT COURSE COMPLETION

After completing this course, you will be able to:

- Implement advanced PowerShell scripting patterns and automation workflows for Windows Server 2025 environments
 - Automate identity and access management for Active Directory, Microsoft Entra ID, and hybrid scenarios
 - Deploy and manage Azure cloud infrastructure and on-premises virtualization platforms programmatically
 - Configure server infrastructure using Desired State Configuration and automate software deployment and patch management
 - Implement secure remote administration, file system management, and SQL Server 2025 database operations through automation
 - Automate network configuration, security monitoring, and compliance reporting for enterprise environments
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PREREQUISITES

- Experience using PowerShell in a Server environment
- Basic PowerShell scripting skills
- Familiarity with Microsoft Azure cloud environment

- Basic understanding of databases
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COURSE OUTLINE

MODULE 1 **Advanced PowerShell Scripting Techniques**

105 min

This module covers advanced scripting patterns, error handling, logging, and modern PowerShell features introduced in PowerShell 7+ and Windows Server 2025.

LESSONS

- Lesson 1:** PowerShell 7+ Evolution and Enterprise Readiness
- Lesson 2:** Modern Syntax Features and Language Enhancements
- Lesson 3:** Advanced Function Design with Parameter Sets
- Lesson 4:** Comprehensive Error Handling Strategies
- Lesson 5:** Enterprise Logging and Transcript Management
- Lesson 6:** Script Modularization and Code Reusability

LAB EXERCISES

- Building a Robust Server Health Monitoring Module with Advanced PowerShell Techniques

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Implement advanced error handling using try-catch-finally blocks and custom error types
- Create reusable functions with parameter validation and pipeline support
- Implement structured logging and transcript mechanisms for audit trails
- Apply modern PowerShell 7+ features including ternary operators and pipeline enhancements

MODULE 2 Identity Management with Active Directory PowerShell

105 min

This module focuses on automating Active Directory administration tasks using PowerShell cmdlets for user, group, and organizational unit management.

LESSONS

- Lesson 1:** Active Directory PowerShell Module Overview
- Lesson 2:** User Account Lifecycle Automation
- Lesson 3:** Group Management and Nested Group Operations
- Lesson 4:** Organizational Unit Structure and Delegation
- Lesson 5:** Active Directory Queries and LDAP Filtering
- Lesson 6:** Password Policy Management and Account Security

LAB EXERCISES

- Automating User Provisioning and Group Management in Active Directory

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Automate user provisioning and deprovisioning workflows in Active Directory
- Manage group memberships and group policy assignments programmatically
- Query Active Directory using LDAP filters and PowerShell cmdlets
- Implement bulk operations for user account management and reporting

MODULE 3 Microsoft Entra ID Automation and Management

105 min

This module covers automating Microsoft Entra ID (formerly Azure AD) administration including user management, group operations, and conditional access policies.

LESSONS

Lesson 1: Microsoft Graph PowerShell SDK Overview

Lesson 2: Authentication and Authorization for Entra ID Automation

Lesson 3: User and Guest Account Management

Lesson 4: Group Creation and Dynamic Membership Rules

Lesson 5: RBAC Role Assignments and Privileged Identity Management

Lesson 6: Conditional Access Policy Automation

LAB EXERCISES

Automating User Lifecycle and RBAC in Microsoft Entra ID

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Connect to Microsoft Entra ID using PowerShell modules and authentication methods
Automate user and group lifecycle management in Entra ID
Implement role-based access control (RBAC) assignments programmatically
Configure conditional access policies and security settings through automation

This module focuses on automating Azure resource deployment and management using PowerShell Az modules and infrastructure-as-code principles.

LESSONS

Lesson 1: Azure PowerShell Az Module Architecture

Lesson 2: Azure Authentication and Subscription Management

Lesson 3: Resource Group and Resource Deployment Automation

Lesson 4: Virtual Machine Provisioning and Scaling

Lesson 5: Virtual Network Configuration and Network Security Groups

Lesson 6: Storage Account Management and Blob Operations

Lesson 7: Azure Policy and Governance Automation

Lesson 8: Cost Management and Resource Tagging

LAB EXERCISES

Automating Azure Infrastructure Deployment with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Deploy and manage Azure resources using PowerShell Az module cmdlets

Implement Azure resource groups, tagging strategies, and governance policies

Automate virtual machine provisioning and configuration in Azure

Manage Azure storage accounts, virtual networks, and related infrastructure

MODULE 5 Desired State Configuration and Server Management 105 min

This module covers implementing Desired State Configuration (DSC) for automated server configuration management and drift detection in Windows Server 2025.

LESSONS

- Lesson 1:** Desired State Configuration Architecture and Components
- Lesson 2:** DSC Configuration Authoring and Compilation
- Lesson 3:** Built-in and Custom DSC Resources
- Lesson 4:** Push and Pull Configuration Delivery Modes
- Lesson 5:** Local Configuration Manager Settings and Configuration Drift Detection
- Lesson 6:** Windows Server 2025 DSC Enhancements and Azure Integration

LAB EXERCISES

- Implementing DSC for Automated Server Configuration Management

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and apply DSC configurations for Windows Server 2025 infrastructure
- Implement DSC resources for role installation, service management, and configuration enforcement
- Configure DSC pull server architecture for centralized configuration management
- Implement configuration drift detection and remediation workflows

MODULE 6 Hyper-V Virtualization Management with PowerShell 105 min

This module focuses on automating Hyper-V virtual machine management, including VM provisioning, snapshot management, and virtual networking configuration.

LESSONS

- Lesson 1:** Hyper-V PowerShell Module Overview
- Lesson 2:** Virtual Machine Lifecycle Automation
- Lesson 3:** Virtual Hard Disk Management and Operations
- Lesson 4:** Checkpoint and Snapshot Management
- Lesson 5:** Virtual Switch Configuration and Network Isolation
- Lesson 6:** VM Replication, Live Migration, and High Availability

LAB EXERCISES

- Automating Hyper-V Virtual Machine Deployment and Network Configuration

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Automate virtual machine creation and configuration using Hyper-V PowerShell cmdlets
- Manage virtual hard disks, checkpoints, and VM state operations programmatically
- Configure virtual networking including virtual switches and network adapters
- Implement VM migration and high availability automation

MODULE 7 Software Deployment and Patch Management Automation

105 min

This module covers automating software deployment, Windows Update management, and patch compliance reporting using PowerShell.

LESSONS

- Lesson 1:** Overview of Automated Software Deployment and Patch Management
- Lesson 2:** PowerShell PackageManagement and Software Repositories
- Lesson 3:** Windows Update Automation with PSWindowsUpdate
- Lesson 4:** WSUS Integration and Third-Party Package Managers
- Lesson 5:** Software Inventory and License Compliance
- Lesson 6:** Patch Compliance Reporting and Rollback Procedures

LAB EXERCISES

- Automating Software Deployment and Patch Management with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Automate software package deployment using PowerShell package management
- Implement Windows Update automation and patch installation workflows
- Create patch compliance reports and vulnerability assessments
- Configure scheduled maintenance windows and update orchestration

MODULE 8 Remote Administration and Session Management

105 min

This module focuses on PowerShell remoting, remote session management, and secure remote administration techniques for distributed environments.

LESSONS

- Lesson 1:** PowerShell Remoting Architecture Overview
- Lesson 2:** Configuring WinRM and Enabling Remoting
- Lesson 3:** One-to-One and One-to-Many Remoting Patterns
- Lesson 4:** Persistent Sessions and Session Management
- Lesson 5:** Credential Management and Authentication Methods
- Lesson 6:** Just Enough Administration (JEA) Implementation
- Lesson 7:** SSH-Based Remoting in PowerShell 7+
- Lesson 8:** Remote Troubleshooting and Debugging

LAB EXERCISES

- Implementing PowerShell Remoting and JEA for Secure Delegation

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure PowerShell remoting and session configurations for secure remote access
- Execute commands across multiple remote systems simultaneously using remoting
- Implement persistent sessions and session management best practices
- Configure JEA (Just Enough Administration) endpoints for delegated administration

MODULE 9 File Systems, Storage, and Data Management

105 min

This module covers automating file system operations, storage management, and data lifecycle tasks using PowerShell on Windows Server 2025.

LESSONS

- Lesson 1:** File System Automation and Path Manipulation
- Lesson 2:** ACL and Permission Management Automation
- Lesson 3:** Advanced File Enumeration and Filtering
- Lesson 4:** Storage Spaces and Storage Pool Management
- Lesson 5:** File Server Resource Manager (FSRM) Automation
- Lesson 6:** Archival and Data Lifecycle Management

LAB EXERCISES

- Automating Storage Management and Data Lifecycle Policies

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Automate file system operations including ACL management and permission auditing
- Implement file archival, retention policies, and data lifecycle management
- Manage Storage Spaces and storage pool configuration programmatically
- Automate file server resource manager (FSRM) configuration and quota management

MODULE 10 SQL Server 2025 Database Administration with PowerShell

105 min

This module focuses on automating SQL Server 2025 database administration tasks including backup operations, user management, and database maintenance.

LESSONS

- Lesson 1:** SQL Server PowerShell
Module Overview
- Lesson 2:** Database Connection
Management and
Authentication
- Lesson 3:** T-SQL Execution and Query
Management
- Lesson 4:** Backup and Restore
Automation
- Lesson 5:** User and Permission
Management Automation
- Lesson 6:** Database Maintenance and
Index Optimization

LAB EXERCISES

- Automating SQL Server 2025 Database
Operations with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Execute SQL queries and stored procedures from PowerShell scripts
- Automate database backup and restore operations for SQL Server 2025
- Manage SQL Server users, logins, and permissions programmatically
- Implement database maintenance plans and health monitoring automation

MODULE 11 Network Configuration and Management Automation

105 min

This module covers automating network configuration tasks including IP addressing, DNS management, firewall configuration, and network troubleshooting.

LESSONS

- Lesson 1:** Network Automation Overview and Strategy
- Lesson 2:** Network Adapter and IP Configuration Automation
- Lesson 3:** DNS Server and Zone Management Automation
- Lesson 4:** Windows Firewall Automation and Security Configuration
- Lesson 5:** Network Diagnostics and Troubleshooting Automation
- Lesson 6:** DHCP and Advanced Network Services Automation

LAB EXERCISES

- Automating Network Configuration and DNS Management

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure network adapter settings and IP addressing through PowerShell
- Automate DNS zone management and record creation
- Implement firewall rule automation and network security configuration
- Perform network diagnostics and troubleshooting using PowerShell cmdlets

MODULE 12 Security Monitoring, Compliance, and Audit Automation

105 min

This module focuses on automating security monitoring, compliance reporting, and audit log analysis for Windows Server 2025 environments.

LESSONS

- Lesson 1:** Security Automation Framework and Architecture
- Lesson 2:** Advanced Audit Policy Configuration and Management
- Lesson 3:** Security Event Log Collection and Analysis
- Lesson 4:** Security Compliance Framework Implementation
- Lesson 5:** Vulnerability Assessment and Remediation Automation
- Lesson 6:** Incident Response Automation and SIEM Integration

LAB EXERCISES

- Automating Security Compliance Audits and Incident Response

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure Windows Security audit policies and event log monitoring automation
- Implement automated security compliance checks and vulnerability scanning
- Create security audit reports and compliance dashboards
- Automate incident response and security remediation workflows

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

- Implement advanced PowerShell scripting patterns including error handling, logging, and parameter validation
- Query and modify Active Directory objects using PowerShell cmdlets
- Automate user provisioning and group membership management in Active Directory
- Connect to Microsoft Entra ID using PowerShell modules
- Create and manage Entra ID users, groups, and role assignments
- Implement conditional access policies and security defaults through automation
- Provision Azure resources using PowerShell Az module cmdlets
- Implement Azure resource tagging and governance policies
- Automate Azure virtual machine deployment and configuration

DAY 2 CAPSTONE Modules 5, 6, 7, 8

SKILLS TO BE INTEGRATED

- Create and apply Desired State Configuration (DSC) resources for server configuration
- Implement DSC pull server architecture and configuration drift detection
- Manage Windows Server roles and features through DSC automation
- Create and manage Hyper-V virtual machines using PowerShell cmdlets
- Automate virtual machine snapshot management and checkpoint operations
- Configure virtual networking and virtual switch settings programmatically
- Automate software package deployment using PowerShell package management
- Implement Windows Update automation and patch compliance reporting
- Create scheduled tasks for ongoing patch management operations
- Configure PowerShell remoting and session configurations
- Execute commands across multiple remote systems simultaneously
- Implement secure credential management for remote administration

DAY 3 CAPSTONE Modules 9, 10, 11, 12

SKILLS TO BE INTEGRATED

- Automate file system operations including ACL management and permission auditing
- Implement automated file archival and retention policies
- Configure Storage Spaces and storage pool management through PowerShell
- Execute SQL Server database queries and stored procedures from PowerShell
- Automate database backup and restore operations using dbatools module
- Implement SQL Server user and permission management automation
- Configure network adapter settings and IP addressing through PowerShell
- Automate firewall rule creation and network security configuration
- Implement DNS zone management and record automation
- Configure Windows Security audit policies and event log monitoring
- Automate security compliance checks and reporting
- Implement just-in-time access and privilege escalation controls

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.